

Exponential Functions & Models

SUPPORTING SKILLS LEGEND

Student learning objectives for this unit

I — Introduce

D — Deepen

A — Apply

R — Reinforce

- ☐ 1. I can **distinguish** linear change from exponential change using differences, ratios, or percent change.

RELATED SKILLS

- ☐ D Compare functions shown in different representations.
- ☐ D Use constant differences or ratios to distinguish linear and exponential patterns.
- ☐ A Calculate and interpret percent increase or decrease.

- ☐ 2. I can **identify** initial value and growth or decay factor in an exponential model.

RELATED SKILLS

- ☐ I Convert a percent increase or decrease to a growth or decay factor.
- ☐ I Identify initial amount, rate, and time interval in context.
- ☐ I Recall and correctly use exponential-function terms including initial value, growth or decay factor, growth or decay rate, and asymptote.
- ☐ D Explain a model parameter in the language of its context.

- ☐ 3. I can **write** an exponential function from a table, graph, or context.

RELATED SKILLS

- ☐ I Determine an exponential model from tabular values.
- ☐ I Solve for an exponential model using two points.
- ☐ D Connect parameters in a formula to features of its graph.
- ☐ D Identify initial amount, rate, and time interval in context.

- ☐ 4. I can **evaluate and interpret** an exponential model.

RELATED SKILLS

- ☐ D Explain a model parameter in the language of its context.
- ☐ D Identify initial amount, rate, and time interval in context.
- ☐ A Attach and interpret units for rates, inputs, and outputs.
- ☐ A Identify input and output values in a formula, graph, table, or context.

- ☐ 5. I can **calculate** growth under periodic compounding.

RELATED SKILLS

- ☐ I Substitute correctly into a periodic-compounding model.
- ☐ A Calculate a percentage of a quantity or determine what percent one quantity is of another.
- ☐ A Convert a percent increase or decrease to a growth or decay factor.

☐ **6. I can **calculate** growth under continuous compounding.**

RELATED SKILLS

- ☐ I Substitute correctly into a continuous-compounding model.
 - ☐ A Identify initial amount, rate, and time interval in context.
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☐ **7. I can **determine** an exponential formula from two points or other sufficient data.**

RELATED SKILLS

- ☐ D Solve for an exponential model using two points.
 - ☐ A Interpret numerator and denominator roles in a rational exponent.
 - ☐ A Isolate a specified variable in an equation or formula.
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☐ **8. I can **determine** domain, range, intercepts, and asymptotes of an exponential function.**

RELATED SKILLS

- ☐ I Identify horizontal asymptotic behavior of an exponential graph.
 - ☐ D Connect parameters in a formula to features of its graph.
 - ☐ D Determine domain restrictions from denominators, even roots, and logarithms.
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☐ **9. I can **compare** the growth rates of two exponential functions.**

RELATED SKILLS

- ☐ I Compare exponential growth rates expressed in different forms.
 - ☐ D Compare functions shown in different representations.
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☐ **10. I can **determine** the percent growth or decay represented by a continuous exponential model.**

RELATED SKILLS

- ☐ I Convert between equivalent discrete- and continuous-growth forms.
 - ☐ D Convert a percent increase or decrease to a growth or decay factor.
 - ☐ A Calculate and interpret percent increase or decrease.
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